



First Semester 2026-2027
Course Handout (Part II)

Date: 06/08/2026

In addition to Part-I (General Handout for all courses appended to the time table) this portion gives further details regarding the course.

Course No. : CE G619
Course Title : Finite Element Analysis
Instructor-in-charge : SHUVENDU NARAYAN PATEL

1. Course Description:

Fundamentals of the finite element method (FEM); basic formulations of FEM; steps in FEM; 1-D, 2-D and 3-D elements; axi-symmetric elements, isoparametric formulations; programming aspects in FEM; plate bending elements; shell elements; material and geometric nonlinearity; Buckling and Vibration problems, application of FEM to civil engineering problems.

2. Scope and Objective of the Course:

Finite element method is the most powerful numerical method widely used for solving problems in different branches of engineering especially Civil Engineering. This method is usually used to solve complex problems with regard to complexity in material, geometry, boundary, and loading conditions, such as non-homogeneous material, irregular geometry, boundary conditions, and static & dynamic loading conditions. The finite element formulation from the weak form of the governing differential equations is also covered. The course is aimed to enable students to understand the advanced concept of finite element method and its application to Civil Engineering.

3. Text Book

R. D. Cook, D. Malkus, M. E. Plesha, and R. J. Witt, Concepts and Applications of Finite Element Analysis, Wiley India, Fourth Edition, 2003.

4. Reference Books

1. O. C. Zienkiewicz, R. L. Taylor and J. Z. Zhu, The Finite Element Method: Its Basis & Fundamentals, Butterworth-Heinemann, 6th edition, Elsevier India, New Delhi, 2005.
2. Huebner and Thornton, The Finite Element Methods for Engineers, 2nd Edition, Mc-Graw-Hill.
3. Chandrupatla and Belegundu, Introduction to Finite Elements in Engineering, 2nd Edition, Prentice-Hall
4. C.S. Krishnamurthy, Finite Element Analysis: Theory and programming, Second Edition, 1994.
5. C. S. Desai and J.F. Abel, Introduction to Finite Element Method, CBS Publishers & Distributors, New Delhi, 1972.
6. Reddy, J.N., "An Introduction to Finite Element Method, 3rd Edition McGraw-Hill.





5. Course Plan:

Module Number	Lecture session/Tutorial Session.	Reference	Learning Outcome
1. Introduction	L1.1. Basic overall introduction in structural mechanics, Problems not under structural mechanics(fluid flow, heat transfer etc.)	[Text book, Ch-1] [RB6, Ch-1]	One can learn how versatile the FEM is, how FEM is used to simulate the natural processes in mathematical form for structural mechanic and non structural mechanics systems.
	L1.2. Need of FEM in solids and structures, basic concept of FEM, advantages of FEM, Steps in FEM, Concepts of compatibility and completeness		
	L1.3. Functionals and Governing Differential Equation, Strong and Weak form of Differential Equations,		
	L1.4. Essential and Natural Boundary Conditions. Finite element formulations(Differential equation based and Non differential equation based).		





2. Variational Formulation and Approximation	L2. 1-5 General introduction, Some mathematical concept and Formulae, Elements of calculus of variation, Integral formulations, Variational methods.	Ch-4, 5 in text book and [Ch-2 in RB6]	One can learn how to develop the weak form of a differential equation. One can understand the variational formulation, classical variational formulation and weak form variational formulation. One can learn Ritz method and Weighted residual methods (like Galerkin and many other). One can also learn modified/weak-form Galerkin methods to solve the governing boundary value problems by transforming the differential equations to a set of algebraic equations.
3. Interpolation function	L3. 1-2 Interpolation function for basic and higher order elements, criterion for selecting the interpolation function.	Ch-3, Text Book	One can learn what is an interpolation function, how to select a function, their properties
4. Generalized Force–Displacement Relationship	L4. 1-2 Principle of virtual work, Derivation of generalized Force-displacement relationship using principle of virtual work, expressions for stiffness matrix and load vectors.	Ch-3, Text Book	One can learn how the principle of virtual work is used to derive stiffness matrix for the structural mechanics problems.



5. One-dimensional Elements	L5. 1-4 Development of interpolation function for one-dimensional elements, Development of stiffness matrix for Bar Element (from weak form of DE and from other forms), Assembly of Elements, properties of structural stiffness matrix, application of boundary conditions, development of load vector, solution of equations, determination of strains, stresses and reactions, analysis of truss structures, Higher order 1-D elements, development of stiffness matrix and load vector for Higher order one-dimensional. Use of this element in other physical problem.	Ch-3, Text Book	One can learn the development of weak form equation of the governing differential equation of one dimensional problems and how to formulate the FE equation to solve.
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6. Two-dimensional Elements	L6. 1-5 Displacement functions for 2-D elements, Stress-strain relation for plane stress and plane strain problems, stiffness matrix for linear triangular element(in structural mechanics-CST) and quadratic triangular (in structural mechanics-LST) element, interpolation function for higher order Triangular elements, Lagrangian and Serendipity shape functions for linear quadrilateral elements and quadratic quadrilateral elements, stiffness matrix of these elements(from weak form of DE and other forms), defects in quadrilateral elements, Comparison of triangular and quadrilateral elements, Use of these element in other physical problem.	Ch-3,Text Book	One can learn the development of weak form equation of the governing differential equation of two dimensional problems and how to formulate the FE equation to solve.
7. Axi-symmetric elements	L7. 1-2 Formulation of axi-symmetric elements and its application.	Ch-14,Text Book	One can learn how to formulate the FE equations for axi-symmetric cases and how to solve those equations.
8. Beam elements	L8 1-3 Stiffness matrix for beam elements(from weak form of DE and other forms), in 2-D and 3-D, shear deformation in beams, analysis of 2-D and 3-D framed structures.	Ch-15,Text Book	One can learn the development of weak form equation of the governing differential equation of beam bending problems and how to formulate the FE equation to solve.





9. 3-D elements	L9 1-2 Review of various 3-D elements, Lagrangian and Serendipity Shape functions for 3-D elements, constitutive matrix for 3-D elasticity	Ch-3,Text Book	One can learn how to formulate the FE equations for 3-D problem cases and how to solve those equations.
10. Isoparametric formulations	L10 1-4 Introduction to isoparametric formulation of different elements, linear quadrilateral elements, Quadratic quadrilateral elements, Hexahedral elements, Incompatible modes, static condensation, choice of numerical integration, consideration to various types of loads, stress calculation, effect of element geometry, validity of elements with iso-parametric formulations, Patch Test, Iso-parametric formulations of Triangular and tetrahedral elements.	Ch-6 and 7,Text Book	One can learn the isoparametric formulation of different elements from the weak form equation development and how to solve those.
11. Plate Bending Elements	L11 1-4 Plate Behaviour, Kirchhoff Plate Elements(from weak form of DE and other forms), Mindlin Plate Elements(from weak form of DE and other forms), shear locking,	Ch-15,Text Book	One can learn the development of weak form equation of the governing differential equations of plate bending problems and how to formulate the FE equation to solve.



12. Shells elements	L12 1-4 Shell theory, Arch Elements, Shells of revolution, general three and four node shell elements, curved isoparametric shell elements, shear and membrane locking.	Ch-16, Text Book	One can learn the development of weak form equation of the governing differential equations of shell bending problems and how to formulate the FE equation to solve.
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6. Evaluation Scheme:

Evaluation Component No	Evaluation Component	Duration	Weightage	Date and time	Nature of component.
1	Mid-Semester Test	1-hour and 30-minutes	20%	As per timetable	CB
2	Project, Assignments, Computer Programs, Seminars, Take Home Tests, Class Test		35%	Continuous	OB
3	Comprehensive Examination	3-hours	45%	As per timetable	CB

7. Chamber Consultation Hour: To be announced in the class.

8. Notices: All notices concerning the course will be communicated through online mode.

9. Make up policies: Makeup may be given only to the genuine cases provided prior permission is taken.

Instructor-In-Charge
CE G619

